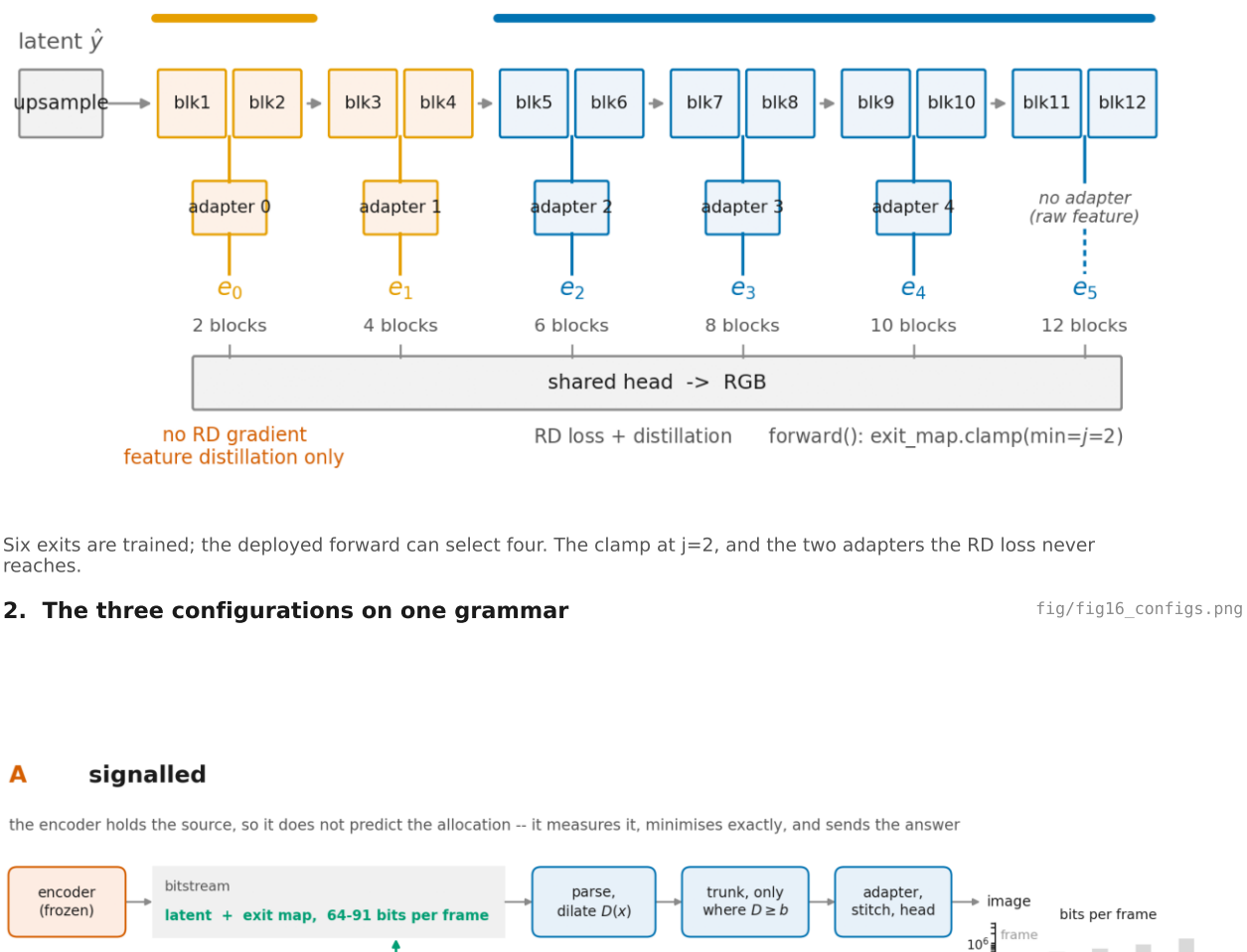


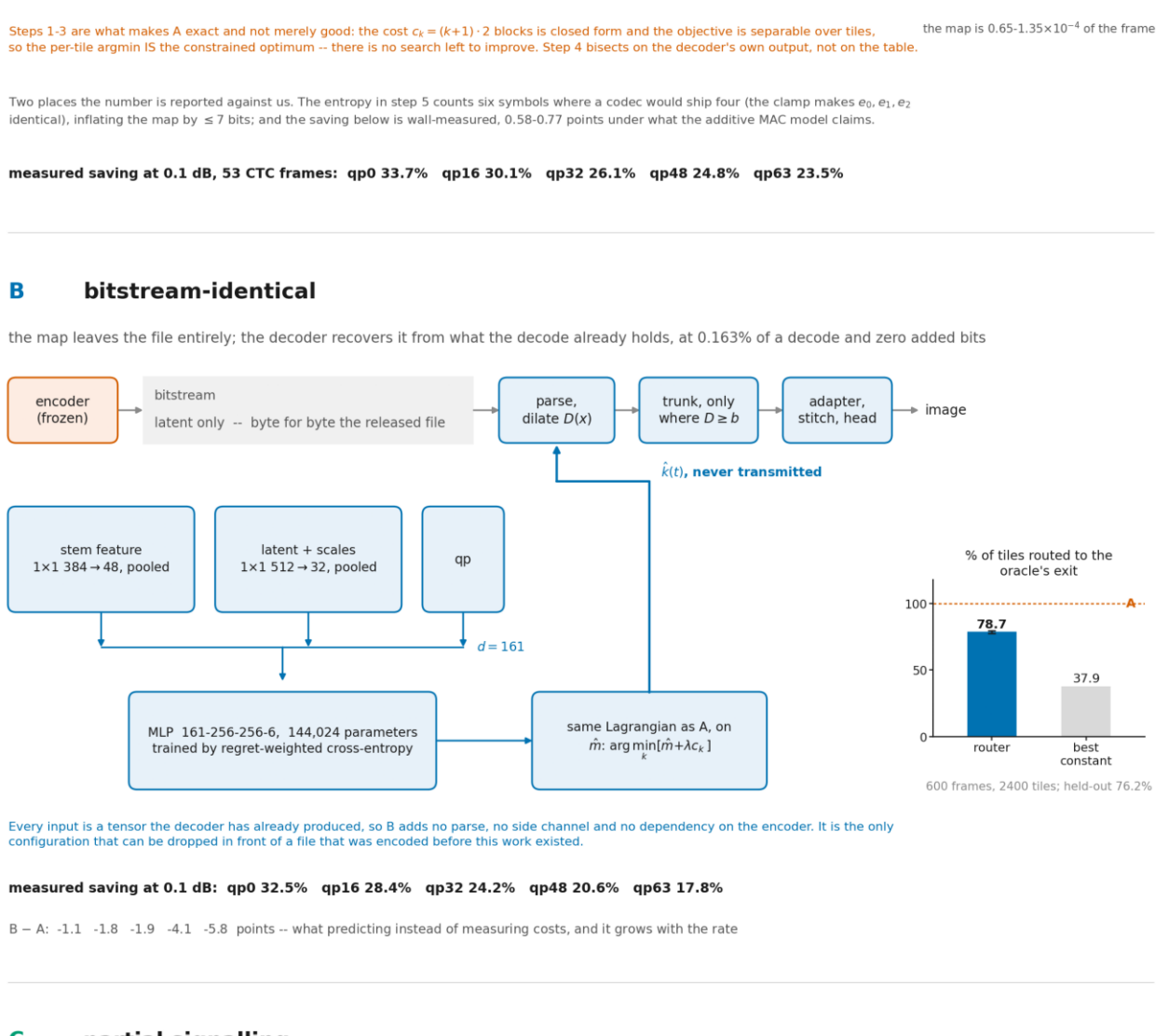
FLEX-UF -- every figure, in the order the argument runs

The ladder the main experiment trains



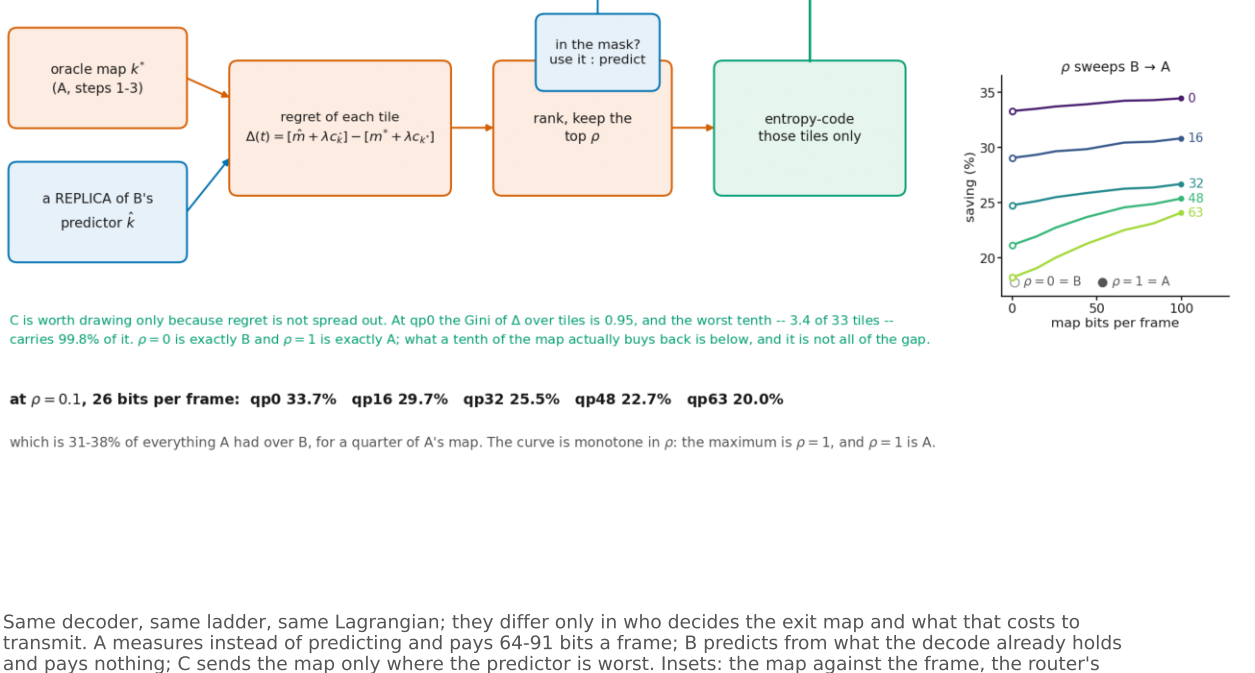
Six exits are trained; the deployed forward can select four. The clamp at $j=2$, and the two adapters the RD loss never reaches.

2. The three configurations on one grammar



measured saving at 0.1 dB, 53 CTC frames: qp0 33.7% qp16 30.1% qp32 26.1% qp48 24.8% qp63 23.5%

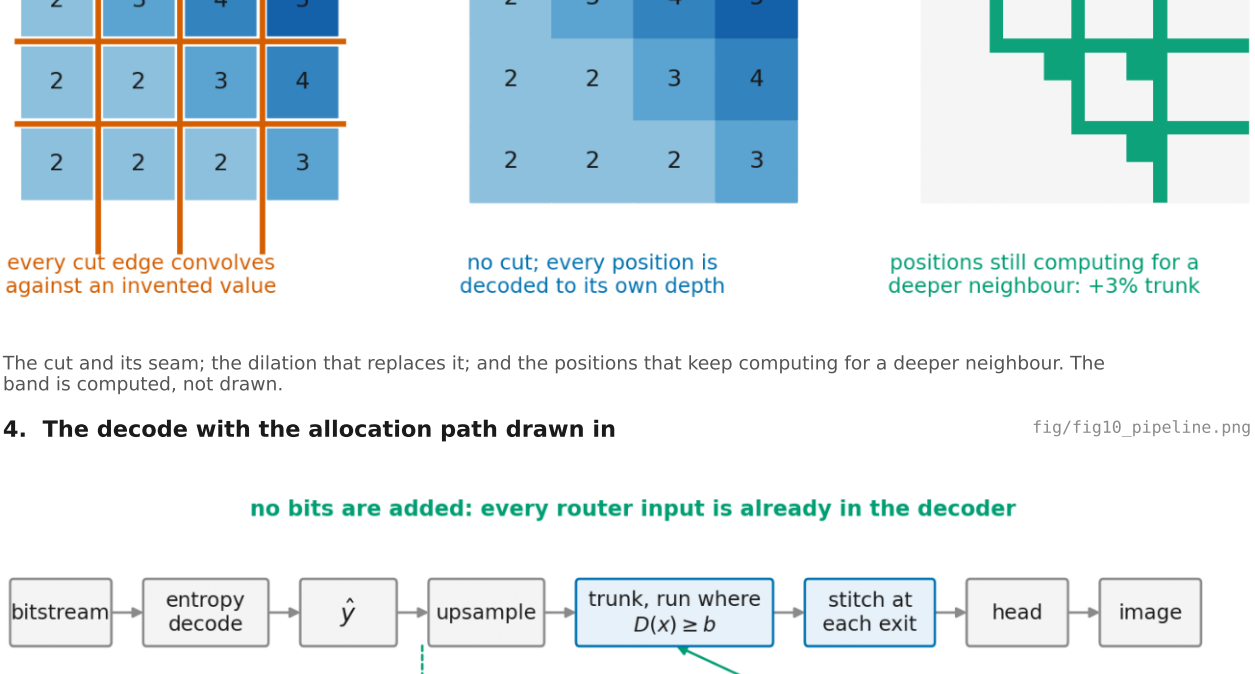
B bitstream-identical



measured saving at 0.1 dB: qp0 32.5% qp16 28.4% qp32 24.2% qp48 20.6% qp63 17.8%

B -- A: -1.1 -1.8 -1.9 -4.1 -5.8 points -- what predicting instead of measuring costs, and it grows with the rate

C partial signalling

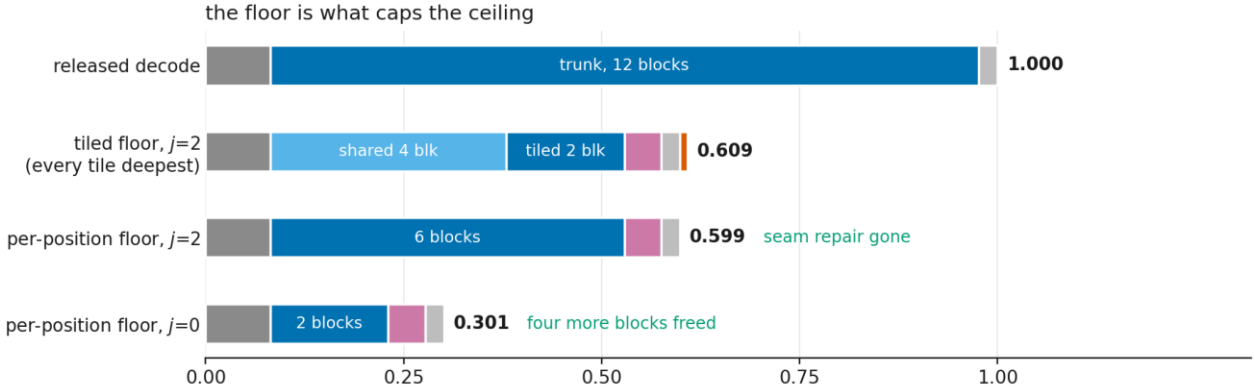


at $\rho=0.1$, 26 bits per frame: qp0 33.7% qp16 29.7% qp32 25.5% qp48 22.7% qp63 20.0%

which is 31.38% of everything A had over B, for a quarter of A's map. The curve is monotone in ρ : the maximum is $\rho=1$, and $\rho=1$ is A.

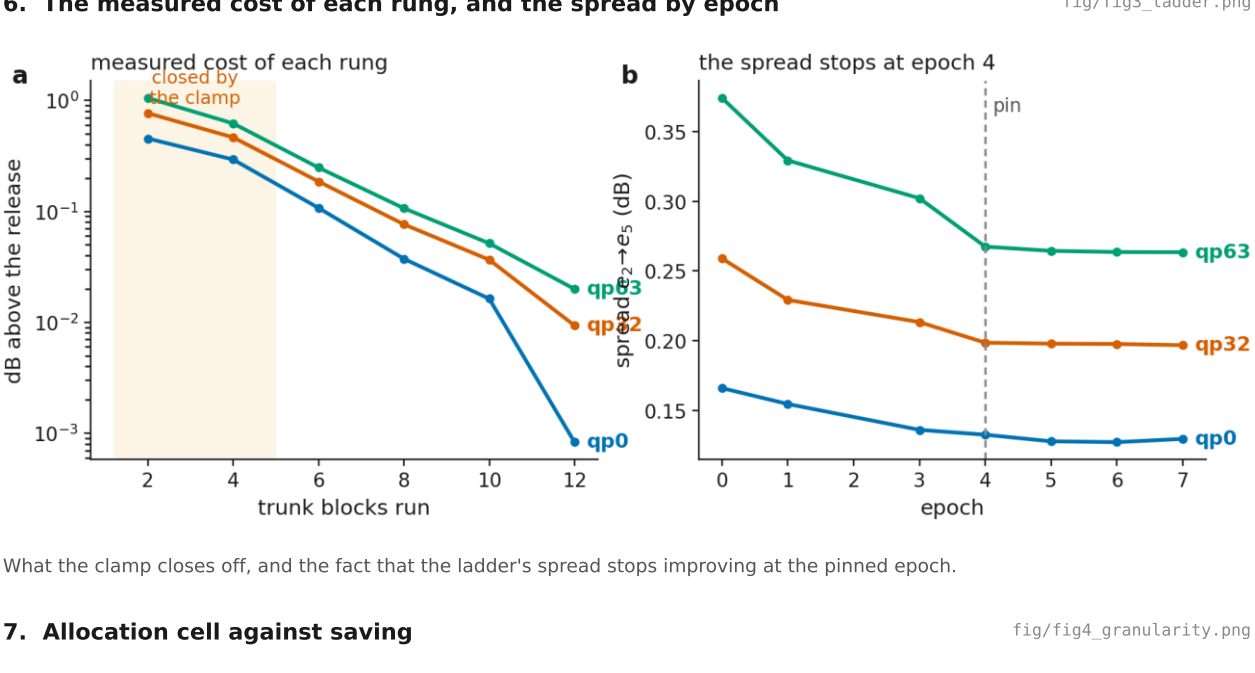
Same decoder, same ladder, same Lagrangian; they differ only in who decides the exit map and what that costs to transmit. A measures instead of predicting and pays 64-91 bits a frame; B predicts from what the decode already holds and pays nothing; C sends the map only where the predictor is worst. Insets: the map against the frame, the router's

3. Tiled decoding, per-position depth, and the band



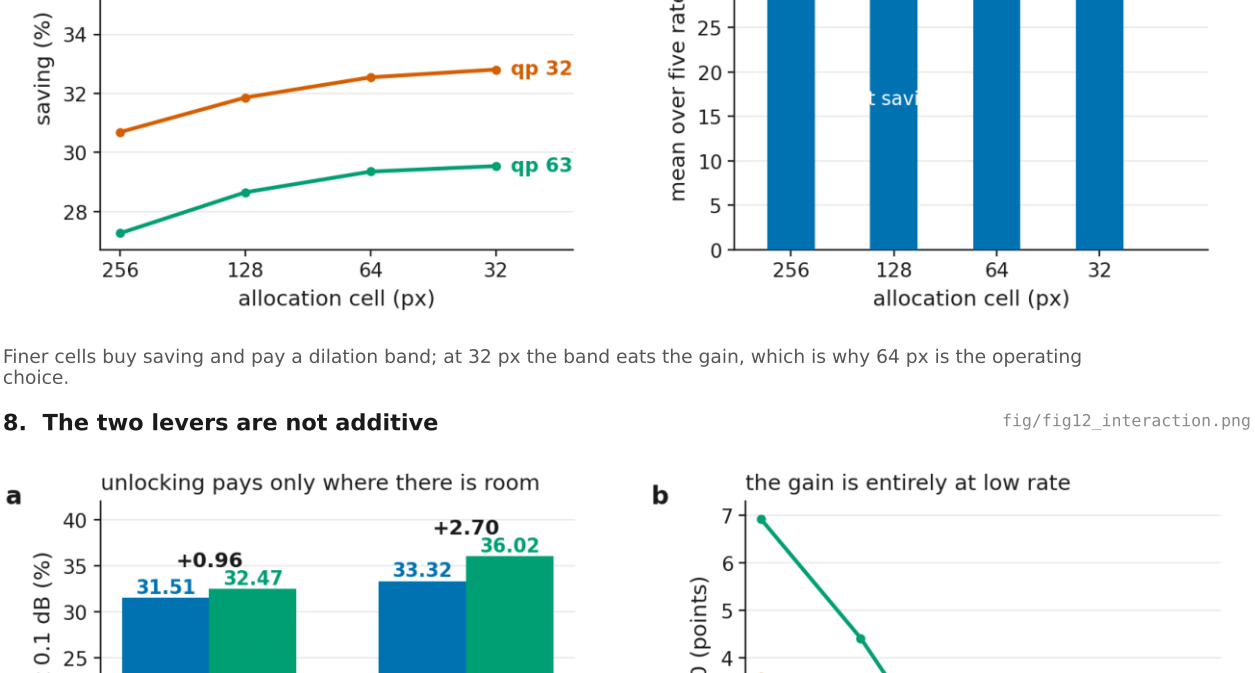
The cut and its seam; the dilation that replaces it; and the positions that keep computing for a deeper neighbour. The band is computed, not drawn.

4. The decode with the allocation path drawn in



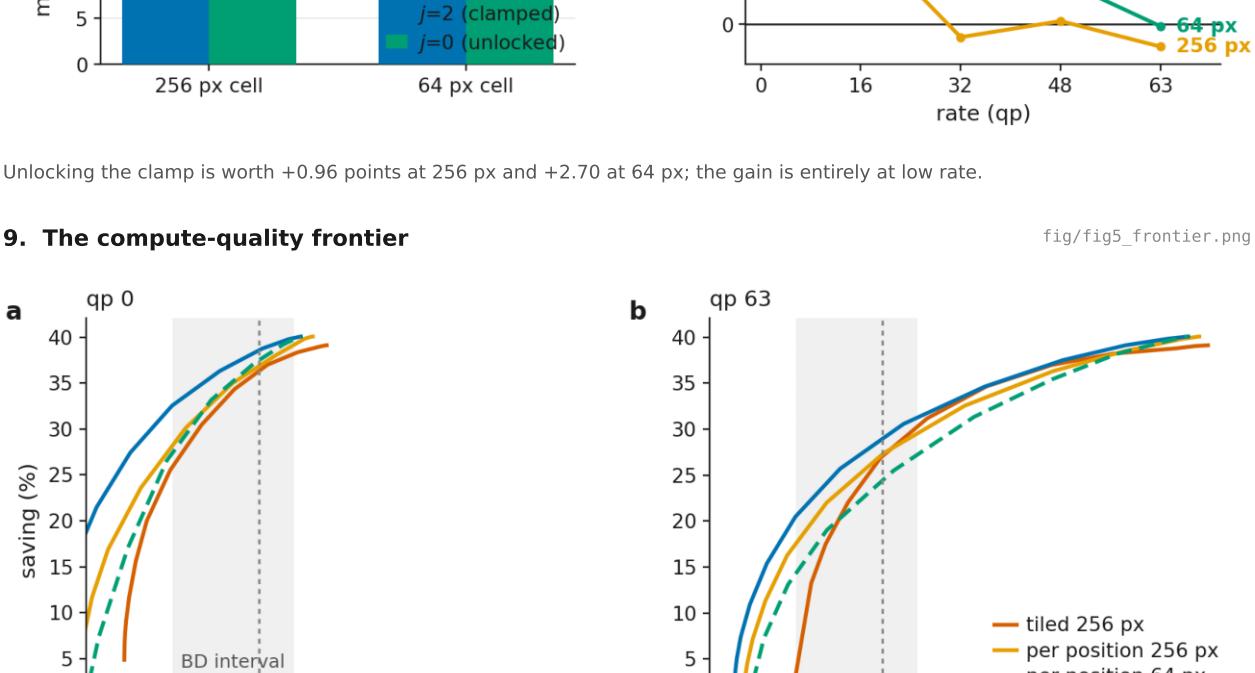
Every router input is already in the decoder, so nothing is added to the bitstream and the map is dilated once before the trunk runs.

5. Where one decode's compute goes



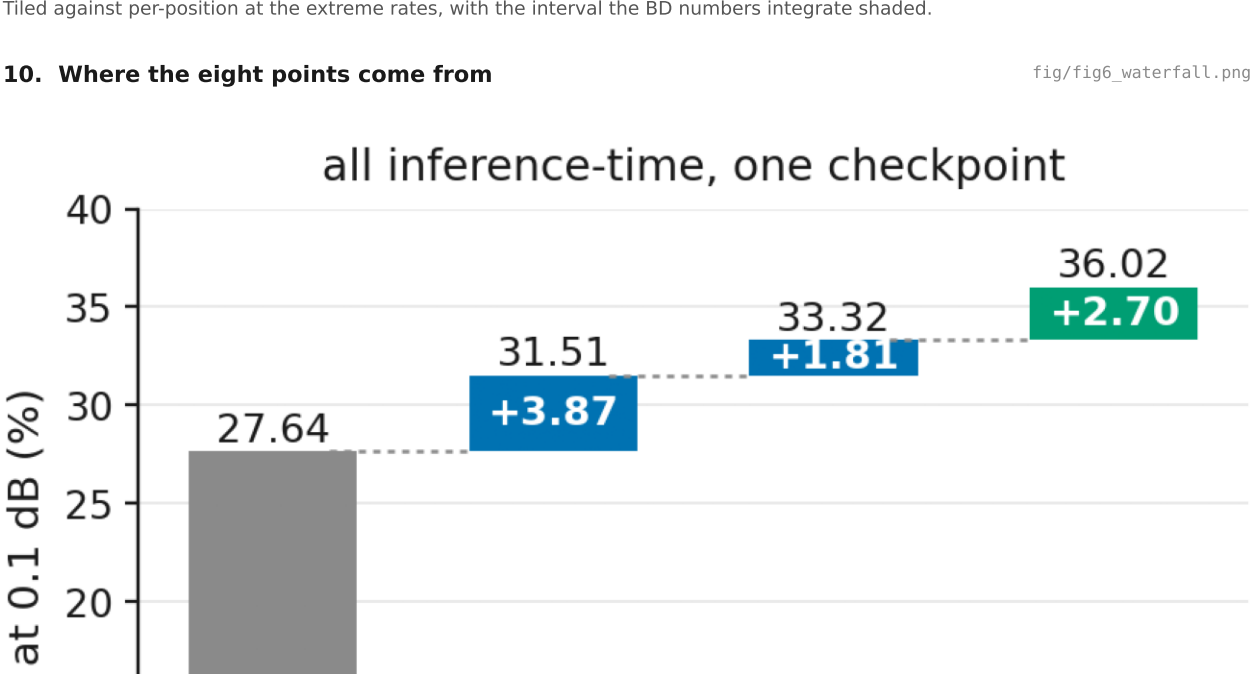
The floor caps the ceiling: 0.609 tiled, 0.599 per-position, 0.301 once the clamp stops forcing four shared blocks on every position.

6. The measured cost of each rung, and the spread by epoch



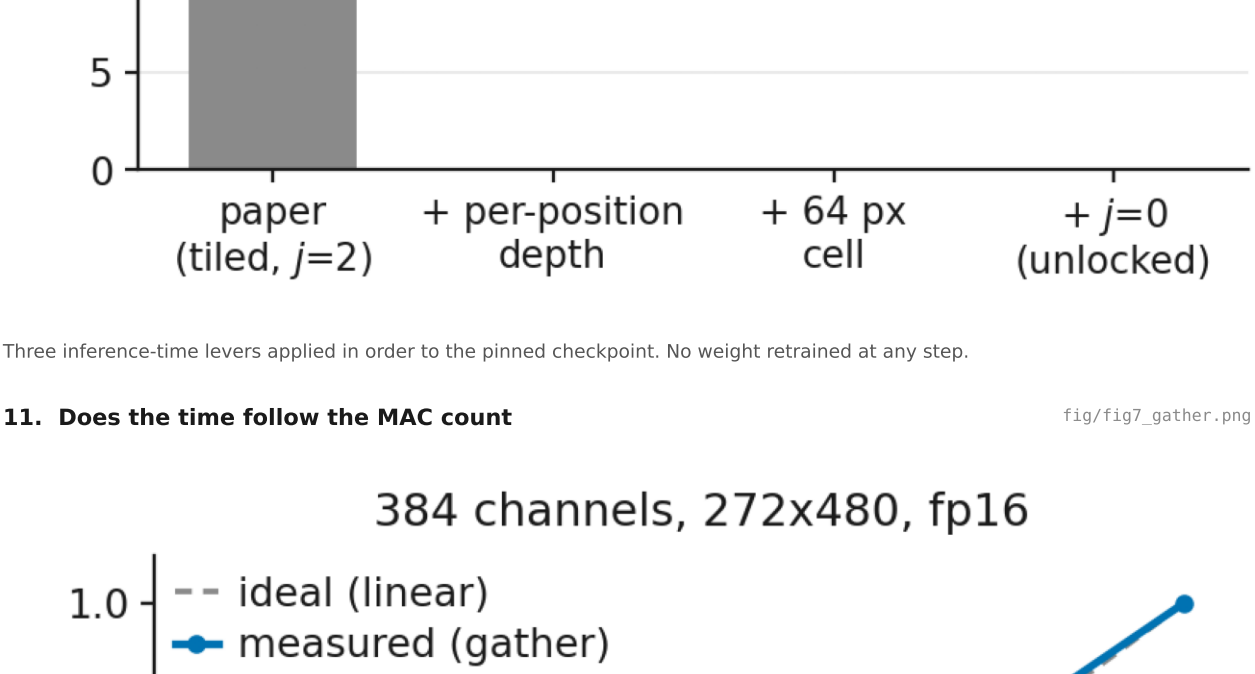
What the clamp closes off, and the fact that the ladder's spread stops improving at the pinned epoch.

7. Allocation cell against saving



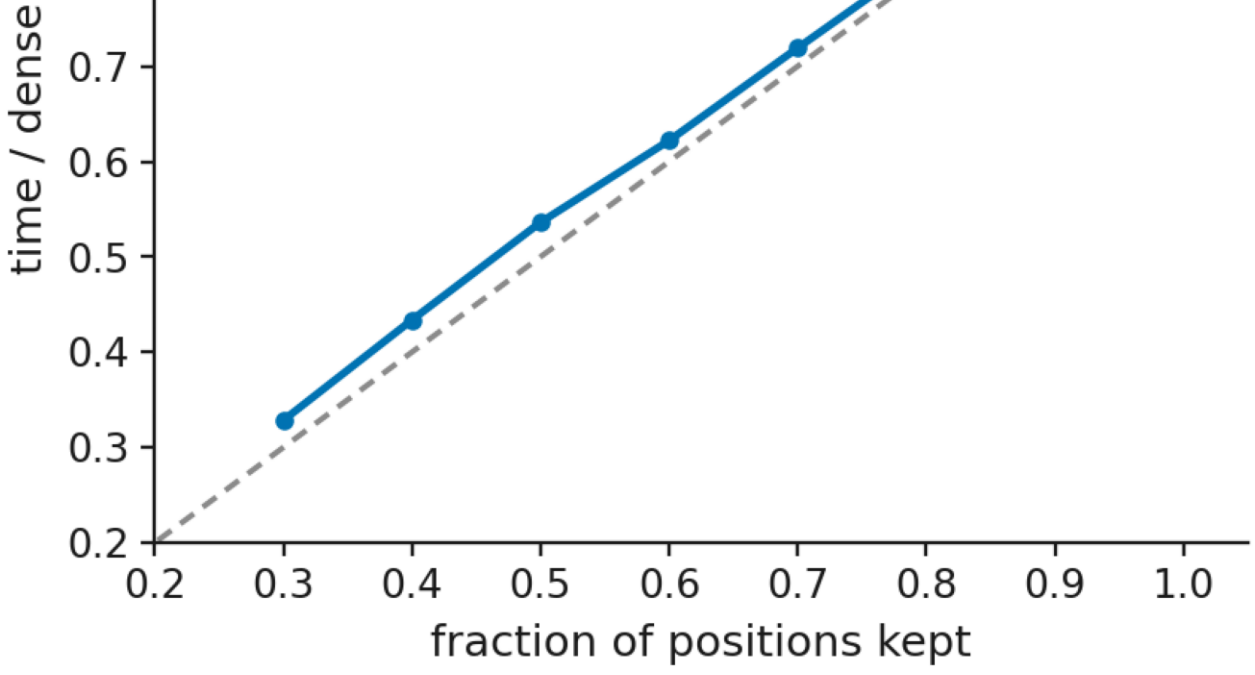
Finer cells buy saving and pay a dilation band; at 32 px the band eats the gain, which is why 64 px is the operating choice.

8. The two levers are not additive



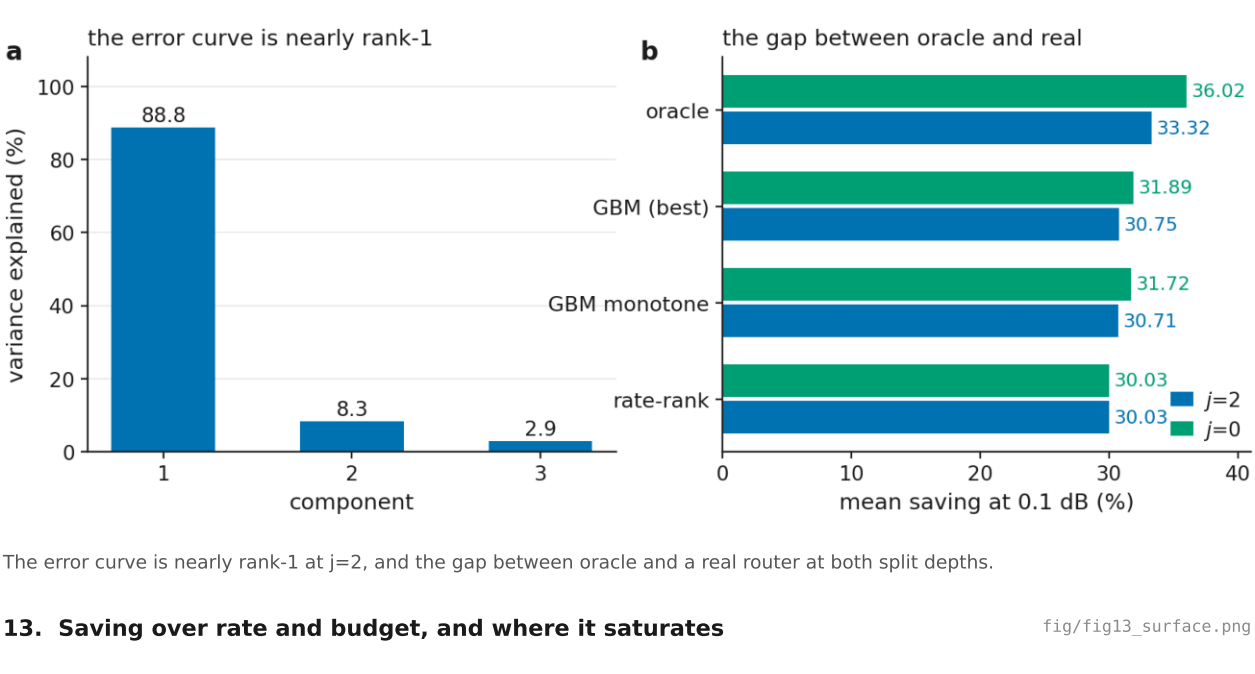
Unlocking the clamp is worth +0.96 points at 256 px and +2.70 at 64 px; the gain is entirely at low rate.

9. The compute-quality frontier



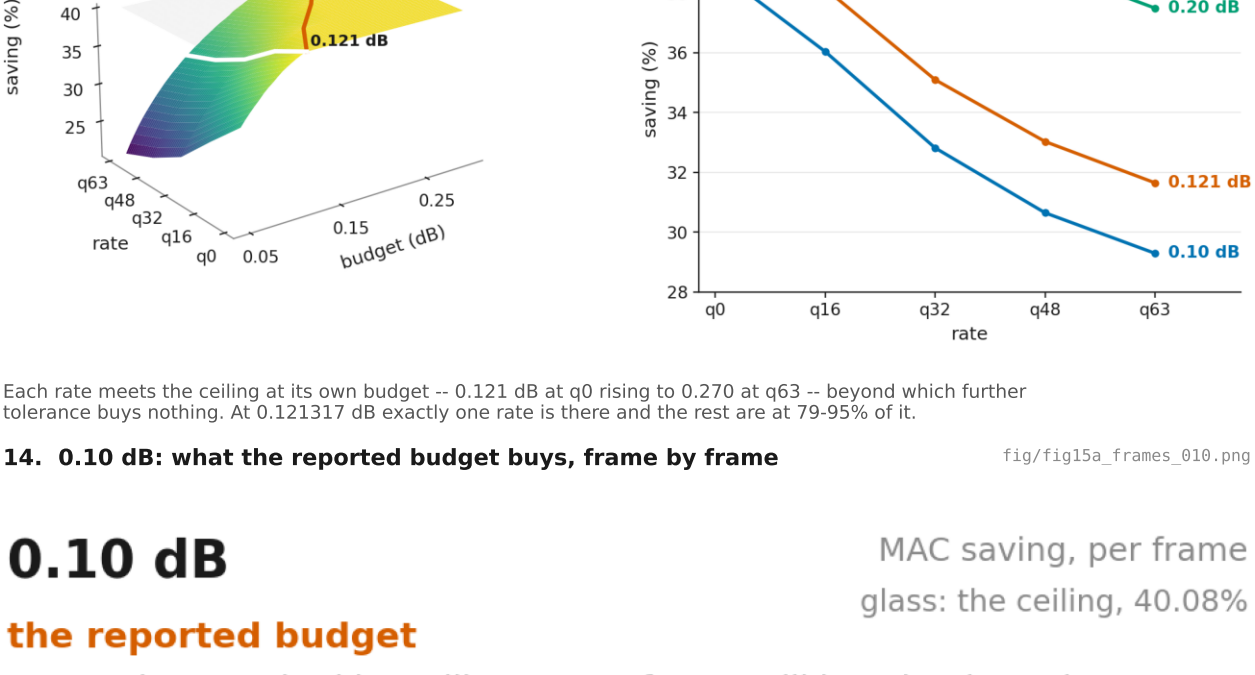
Tiled against per-position at the extreme rates, with the interval the BD numbers integrate shaded.

10. Where the eight points come from



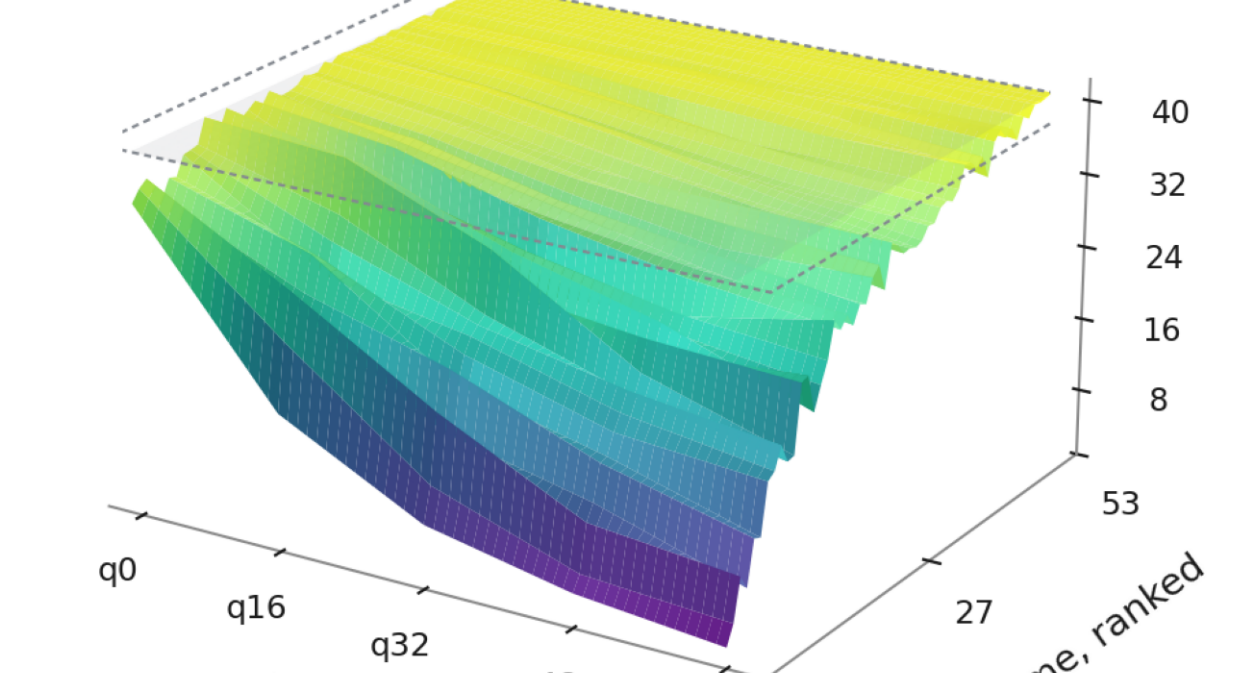
Three inference-time levers applied in order to the pinned checkpoint. No weight retrained at any step.

11. Does the time follow the MAC count



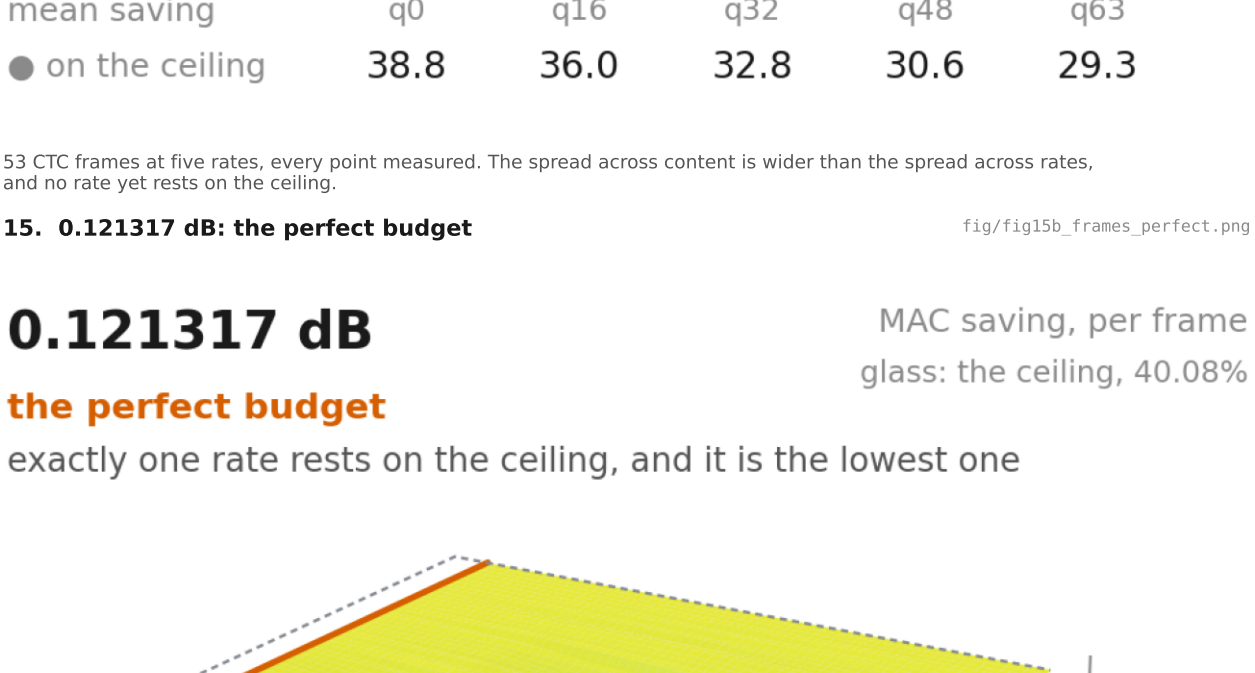
Gather, two pointwise convolutions, scatter, at the decoder's shapes. Time tracks the kept fraction to within 0.008-0.036, with RANDOM indices.

12. The routing axis



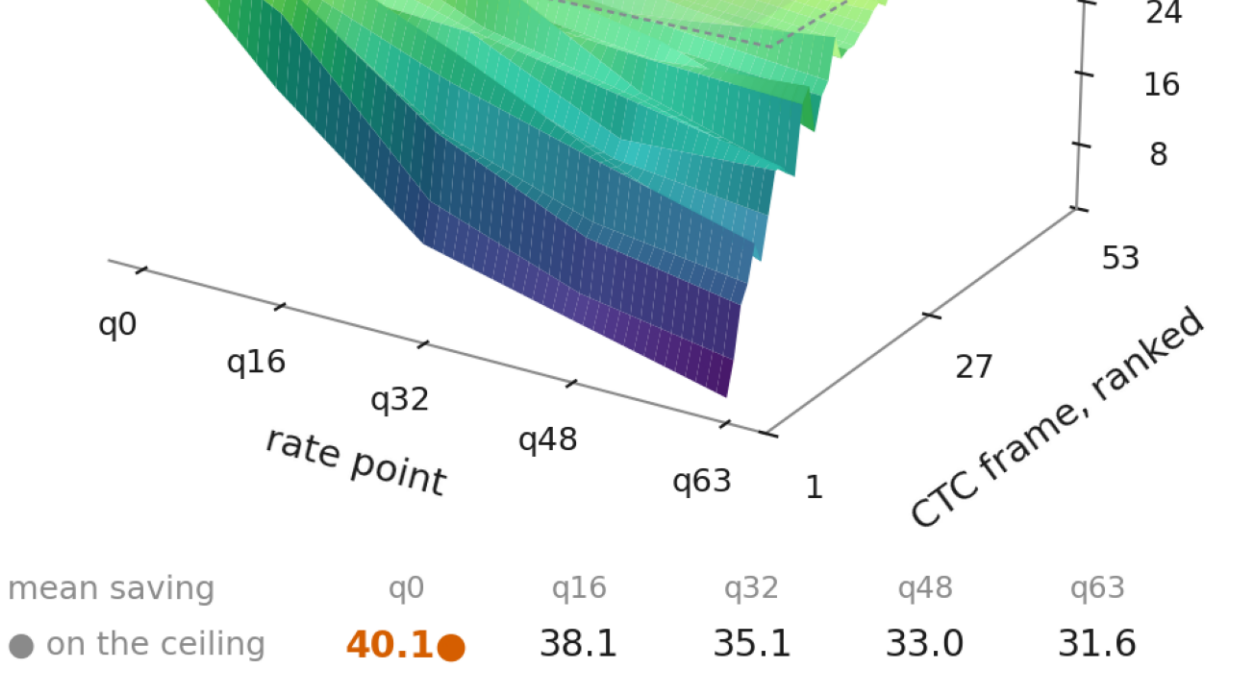
The error curve is nearly rank-1 at $j=2$, and the gap between oracle and a real router at both split depths.

13. Saving over rate and budget, and where it saturates



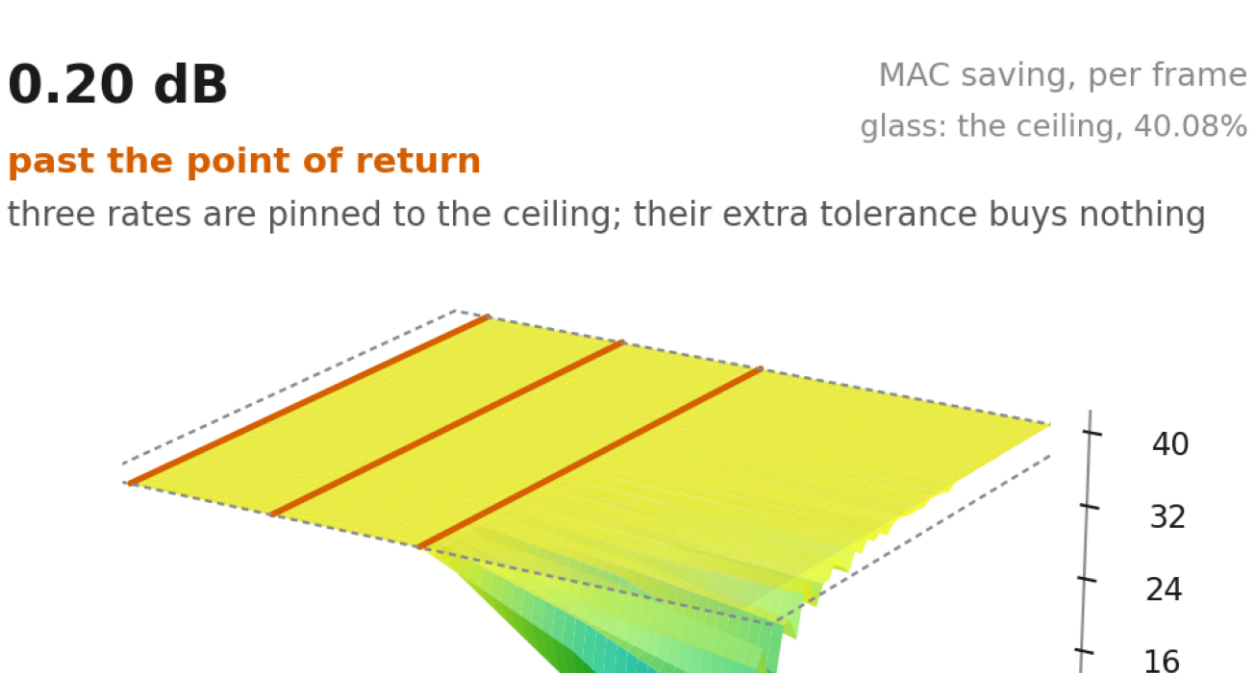
Each rate meets the ceiling at its own budget -- 0.121 dB at q0 rising to 0.270 at q63 -- beyond which further tolerance buys nothing. At 0.121317 dB exactly one rate is there and the rest are at 79-95% of it.

14. 0.10 dB: what the reported budget buys, frame by frame



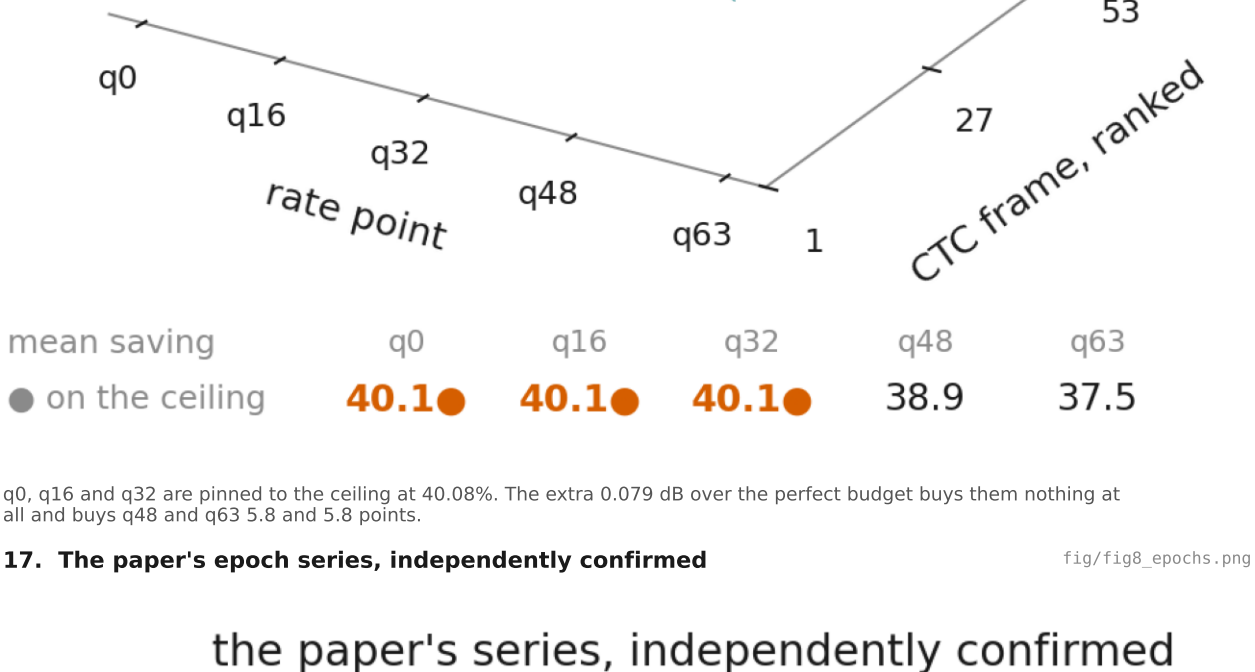
53 CTC frames at five rates, every point measured. The spread across content is wider than the spread across rates, and no rate yet rests on the ceiling.

15. 0.121317 dB: the perfect budget



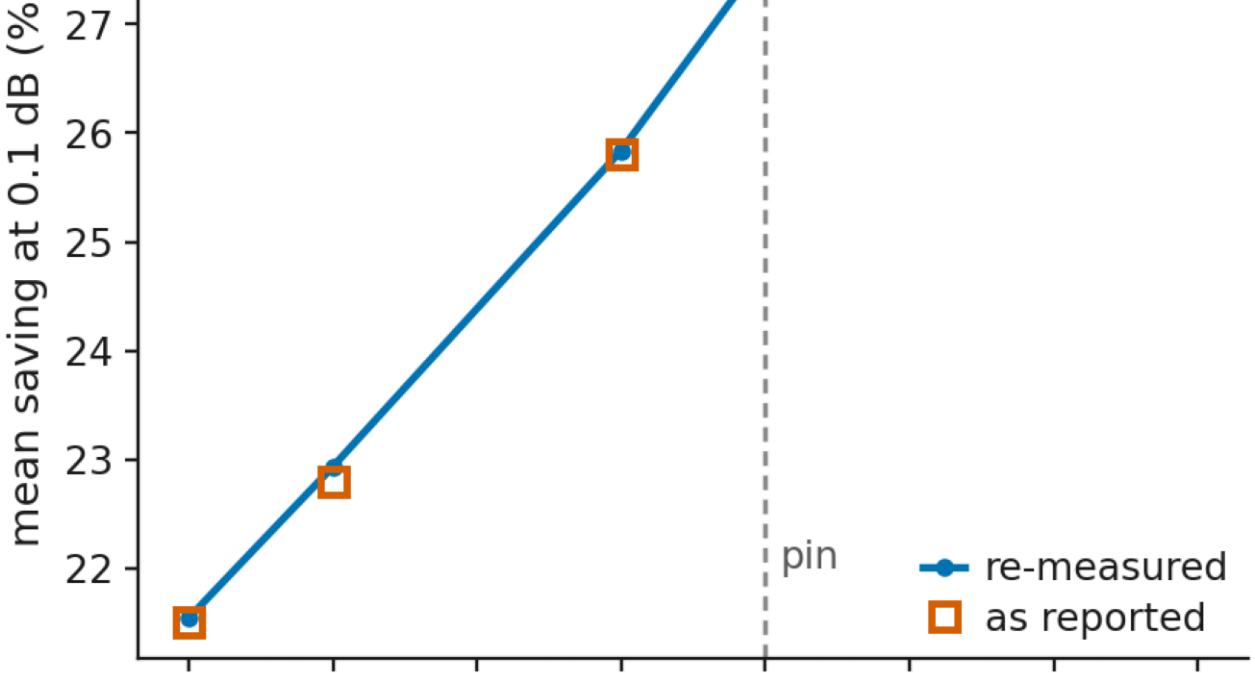
q0's whole row lies on the ceiling and nothing else does. This is the largest budget at which no tolerance is wasted: one rate has run out of depth to give up, the next does not until 0.153 dB.

16. 0.20 dB: past the point of return



q0, q16 and q32 are pinned to the ceiling at 40.08%. The extra 0.079 dB over the perfect budget buys them nothing at all and buys q48 and q63 5.8 and 5.8 points.

17. The paper's epoch series, independently confirmed



Re-measured with current scripts on every surviving checkpoint.